Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/08/2014
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Risk of Intervertebral Disc Damage after Prolonged Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lotz, Jeffrey (University of California, San Francisco) O'Neill, Conor (Self) Sayson, Jojo (Ola Grimsby Institute, San Diego) Chiang, Stephen (Methodist Hospital) Chang, Douglas (University Of California, San Diego) Allon, Moshe (Self) Parazynski, Scott (The Methodist Hospital Research Institute) Riascos-Castaneda, Roy (University of Texas Medical Branch)		
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Performance Goal Text:			

Task Description:	Editor's Note (5/2/2014): NOTE THIS IS A CONTINUATION OF FUNDING FOR NNX10AM18G WITH THE SAME TITLE AND PRINCIPAL INVESTIGATOR. Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (IRP Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crew members before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crew member injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
	We have made significant progress over the past 12 months conducting our pre- and post-flight tests, consenting more ISS crew members, updating IRB applications, submitting amendments and receiving approvals from the UCSD and NASA-JSC Institutional Review Boards (IRBs). Importantly for this year, we performed a second, full set of Pre- and Post-Flight testing sessions for our second crew member for IVD. Three more crew members were tested pre-flight. We were very successful in recruiting more ISS crew members over the past year. To improve our tests and protocol, we amended our JSC-CPHS Protocol 10-072 in March 2014 in order to: 1) Update our personnel, 2) Remove (+1 day) flexibility from the R+1 post flight MRI/MRS, 3) Allow Spinal Kinematics to be scheduled before the MRI/MRS based on UTMB facility constraints, 4) Reduce the amount of weight that can be lifted before pre and post flight MRI/MRS. We also requested that crew members avoid Neutral Buoyancy Laboratory (NBL) runs, acrobatic flying, and provocative movement for 24 hours prior to the session. Provocative movements include rapid movements that place unusual stress on the spine (e.g. the Computerized Dynamic Posture test, Rock Translation test & Jump Down test that occur as a part of Field Test), 5) The changes noted above are also reflected in the NASA Lay Consent for ISS crew members as well as in the Multi-National Consent. Over the past year we have published 7 papers and 2 abstracts: 1: Shymon SJ, Yaszay B, Dwek JR, Proudfoot JA, Donohue M, Hargens AR. Altered disc compression in children with idiopathic low back pain: An upright magnetic resonance imaging backpack study. Spine 2014 Feb;39(3):243-48 [Epub 2013 Nov 18]. DOI: 10.1097/BRS.0000000000000114. PubMed PMID: 24253789. 2: Kim SH, Neuschwander TB, Macias BR, Bachman L Jr, Hargens AR. Upper extremity hemodynamics and sensation with backpack loads. Appl Ergon 2014 May;45(3):608-12 [Epub 2013 Sep 26]. PII: S0003-6870(13)00170-1. DOI: 10.1016/j.apergo.2013.08.005. PubMed PMID: 24075289 3: Schlabs T, Rosales-Velderrain A, Ruckstuhl H, Stahn AC, Hargens AR. Comparison of cardiovascular and biomechanical parameters of supine lower body negative pressure and upright lower body positive pressure to simulate activity in 1/6 G and 3/8 G. J Appl Physiol 2013 Jul;115(2):275-84 [Epub 2013 May 2]. DOI: 10.1152/japplphysiol.00990.2012. PubMed PMID:

Task Progress:

23640597. PubMed Central PMCID: PMC3727004

4: Rodríguez-Soto AE, Jaworski R, Jensen A, Niederberger B, Hargens AR, Frank LR, Kelly KR, Ward SR. Effect of load carriage on lumbar spine kinematics. Spine 2013 Jun;38(13):E783-91. DOI: 10.1097/BRS.0b013e3182913e9f. PubMed PMID: 23524870

5: Mateus J, Hargens AR. Bone hemodynamic responses to changes in external pressure. Bone 2013 Feb;52(2):604-10 [Epub 2012 Nov 17]. DOI: 10.1016/j.bone.2012.11.010. PubMed PMID: 23168293

6: Hargens AR, Bhattacharya R, Schneider SM. Space physiology VI: exercise, artificial gravity, and countermeasure development for prolonged space flight. Eur J Appl Physiol 2013 Sep;113(9):2183-92 [Epub 2012 Oct 19]. DOI: 10.1007/s00421-012-2523-5. PubMed PMID: 23079865

7: Goswami N, Batzel JJ, Clément G, Stein TP, Hargens AR, Sharp MK, Blaber AP, Roma PG, Hinghofer-Szalkay HG. Maximizing information from space data resources: a case for expanding integration across research disciplines. Eur J Appl Physiol 2013 Jul;113(7):1645-54 [Epub 2012 Oct 17]. DOI: 10.1007/s00421-012-2507-5. PubMed PMID: 23073848

8: Sayson JV, Hargens AR, Chang D, Macias B. A novel exercise regimen for preserving cervico-thoracic muscle function during prolonged exposure to microgravity. Aviation Space and Environmental Medicine 2014;85(3):286.

9: Sayson JV, Lotz J, Parazynski SE, Hargens AR. Abstract: Back pain mechanisms in microgravity. Annual Meeting Program and Abstract Issue, Aviation Space and Environmental Medicine 2013;84(4):285-86.

In Review:

Bailey, J.F., Hargens, A.R., Cheng, K.K., and Lotz, J.C., Effect of microgravity on the biomechanical properties of the lumbar and caudal intervertebral discs in mice. Journal of Biomechanics.

In Preparation:

Laws, C., Hargens, A.R., and Lotz, J.C., The effect of simulated microgravity on lumbar biomechanics and annulus fiber strain: an in vitro study. In preparation for Journal of Biomechanics.

Laws, C., Hargens, A.R., Sayson, J.V., and Lotz, J.C., Spinal disc herniation and microgravity, a review of the biomechanical literature. Invited review for Aviation, Space, and Environmental Medicine (ASEM).

Recently, we also made progress as follows:

• Dr. Lotz presented an invited lecture on the IVD study at the Philadelphia New Horizons in Intervertebral Disc Research Symposium on 8 Nov 13.

• Dr. Lotz presented updates on the first two crew members to IVD Flight Study Data Review members at NASA JSC on 19 Dec 13.

• Dr. Hargens presented STS-131 and STS-133 mice disc results in Bone Session at HRP Workshop on 12 Feb 14.

• Dr. Hargens briefed four ISS crewmembers for possible consenting and inclusion as subjects for IVD.

• Dr. Hargens updated the Experiment Document for IVD and amended the NASA IRB accordingly.

• Dr. Hargens renewed the UCSD IRB for the IVD project.

• Dr. Sayson was invited to present an abstract, “Back Pain Mechanisms on the International Space Station: a preliminary report,” J.V. Sayson, J.C. Lotz, R. Riascos, D. G. Chang, S, E. Parazynski, and A. R. Hargens, The American Astronautical Society’s 3rd Annual International Space Station Research and Development Conference to be held at the Hyatt Regency McCormick Place, Chicago, Illinois on 19 June 2014.

• Dr. Sayson delivered the Keynote Address, “Astronaut Low Back Pain in Space, HNPs Post-flight and Proposed Countermeasures,” at the 2013 California Physical Therapy Association Annual Conference at the Pasadena Convention Center, Pasadena, California on 21 Sept. 2013.

• Dr. Sayson presented “Spinal Changes after a 6 month mission on board the International Space Station: a preliminary report,” at the Ola Grimsby Institute Competency Forum at Seattle, Washington on 24 Aug. 2013.

• Dr. Sayson represented the IVD Team in a Poster Presentation “Back Pain Mechanisms in Microgravity,” Sayson JV, Hargens AR, Parazynski SE, and Lotz JC, at the 84th Annual Scientific Meeting of the Aerospace Medical Association at the Sheraton Chicago Hotel & Towers, Chicago, IL, May 12-16, 2013.

• Our IVD team performed pre-flight tests on a 5th crew member at UTMB-Victory Lakes on 2-3 April 2014.

• We established a secured web portal for storing and sharing study data at UCSD through the iDASH (Integrating Data for Analysis, Anonymization and Sharing) resource, supported by the National Centers for Biomedical Computing (NCBC) under the NIH Roadmap for Bioinformatics and Computational Biology.

• Initial post-flight observations: 1) Increased lumbar IVD heights as demonstrated by morphological measures of the supine MRI scans, 2) Increased lumbar IVD compressibility with upright MRI scans, 3) Increased bending stiffness using Spinal Kinematics testing, and 4) Increased low back pain using the visual-analog scale.

• Supplementary observations from human cadaveric testing: 1) disc swelling increases disc height and reduces lumbar lordosis;

	2) spine bending stiffness increases proportionally to swelling-induced increases in disc height; 3) disc swelling increases annular strain and nuclear pressure during spine movement. These data quantify the increased risk for disc tissue damage that occurs after simulated microgravity.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Abstracts for Journals and Proceedings	Sayson JV, Hargens AR, Chang D, Macias B. "A novel exercise regimen for preserving cervico-thoracic muscle function during prolonged exposure to microgravity." Presented at the 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):286. See http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 , Mar-2014
Articles in Peer-reviewed Journals	Shymon SJ, Yaszay B, Dwek JR, Proudfoot JA, Donohue M, Hargens AR. "Altered disc compression in children with idiopathic low back pain: An upright magnetic resonance imaging backpack study." (Phila Pa 1976). 2014 Feb 1;39(3):243-8. [Epub 2013 Nov 18]. http://dx.doi.org/10.1097/BRS.0000000000000114 ; PubMed PMID: 24253789 ; PubMed Central PMCID: PMC3947616 , Feb-2014
Articles in Peer-reviewed Journals	Kim SH, Neuschwander TB, Macias BR, Bachman L Jr, Hargens AR. "Upper extremity hemodynamics and sensation with backpack loads." Appl Ergon. 2014 May;45(3):608-12. Epub 2013 Sep 26. http://dx.doi.org/10.1016/j.apergo.2013.08.005 ; PubMed PMID: 24075289 , May-2014
Articles in Peer-reviewed Journals	Schlabs T, Rosales-Velderrain A, Ruckstuhl H, Stahn AC, Hargens AR. "Comparison of cardiovascular and biomechanical parameters of supine lower body negative pressure and upright lower body positive pressure to simulate activity in 1/6 G and 3/8 G." J Appl Physiol (1985). 2013 Jul 15;115(2):275-84. Epub 2013 May 2. http://dx.doi.org/10.1152/japplphysiol.00990.2012 ; PubMed PMID: 23640597; PubMed Central PMCID: PMC3727004 , Jul-2013
Articles in Peer-reviewed Journals	Hargens AR, Bhattacharya R, Schneider SM. "Space physiology VI: exercise, artificial gravity, and countermeasure development for prolonged space flight." Eur J Appl Physiol. 2013 Sep;113(9):2183-92. Epub 2012 Oct 19. http://dx.doi.org/10.1007/s00421-012-2523-5 ; PubMed PMID: 23079865 , Sep-2013
Articles in Peer-reviewed Journals	Goswami N, Batzel JJ, Clément G, Stein TP, Hargens AR, Sharp MK, Blaber AP, Roma PG, Hinghofer-Szalkay HG. "Maximizing information from space data resources: a case for expanding integration across research disciplines." Eur J Appl Physiol. 2013 Jul;113(7):1645-54. Epub 2012 Oct 17. http://dx.doi.org/10.1007/s00421-012-2507-5 ; PubMed PMID: 23073848 , Jul-2013